a coagulating enzyme called **vesiculase** which enhances sperm motility. The bulbourethral glands are inferior to the prostate and their secretions also help in the lubrication of the penis. The prostate encircles the urethra and is just below the urinary bladder and secretes a slightly acidic fluid that contains citrate, several enzymes and prostate specific antigens. **Semen** or seminal fluid is a milky white fluid which contains sperms and the seminal plasma (secreted from the seminal vesicles, prostate gland and the bulbourethal glands). The seminal fluid acts as a transport medium, provides nutrients, contains chemicals that protect and activate the sperms and also facilitate their movement.

The penis is the male external genitalia functioning as a copulatory organ. It is made of a special tissue that helps in the erection of penis to facilitate insemination. The enlarged end of the penis called glans penis is covered by a loose fold of skin called foreskin or prepuce.

The female reproductive system is far more complex than the male because in addition to gamete formation, it has to nurture the developing foetus. The female reproductive system consists of a pair of ovaries along with a pair of oviducts, uterus, cervix, vagina and the external genitalia located in the pelvic region **(Fig. 2.3 a)**. These parts along with the mammary glands are integrated structurally and functionally to support the process of ovulation, Fertilization, pregnancy, child birth and child care.

Ovaries are the primary female sex organs that produce the female gamete, ovum. The ovaries are located one on each side of the lower abdomen. The ovary is an elliptical structure about 2-4 cm long. Each ovary is covered by a thin cuboidal epithelium called the germinal epithelium which encloses the **ovarian stroma**. The stroma is differentiated as the outer cortex and inner medulla. Below the germinal epithelium is a dense connective tissue, the **tunica albuginea**. The cortex appears dense and granular due to the presence of ovarian follicles in various stages of development. The medulla is a loose connective tissue with abundant blood vessels, lymphatic vessels and nerve fibres. The ovary remains attached to the pelvic wall and the uterus by an ovarian ligament called **mesovarium**.

The fallopian tubes (uterine tubes or oviducts), uterus and vagina constitute the female accessory organs **(Fig. 2.3 b)**. Each fallopian tube extends from the periphery of each ovary to the uterus. The proximal part of the fallopian tube bears a funnel shaped **infundibulum**. The edges of the infundibulum have many finger like projections called **fimbriae**

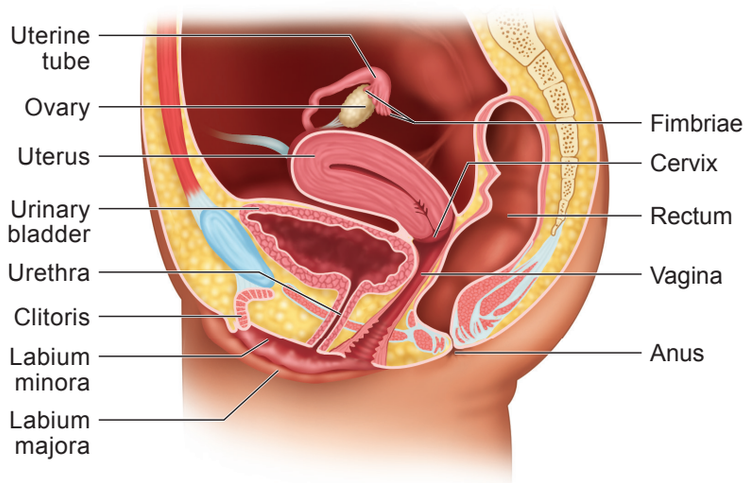


Fig. 2.3 (a) Female pelvis showing reproductive system



Female uterus contains one of the strongest muscles of the human body.

XII Std Biology-Zoology Chapter-2 EM.indd 14 23/12/2022 11:08:24